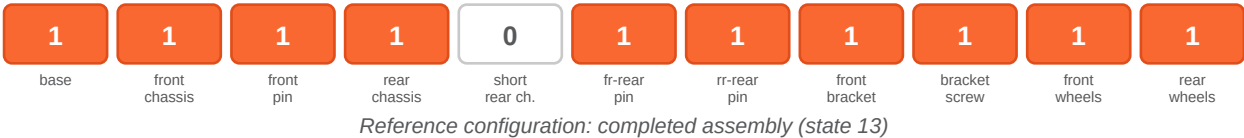


STEMFIE CONSTRUCTION-SET VEHICLE

Assembly and Rear-Chassis Maintenance Manual



Document No.	IR-MAN-001
Revision	B
Status	Released
Reference model	IndustReal dataset / STEMFIE SPS-000001 (open source)
Derived from	IndustReal procedure definitions, assembly-state encodings and 3D part geometries
Applies to	New-build assembly (Procedure A) and rear-chassis upgrade (Procedure B)

Revision history A - initial release | B - tone formalized, photographic placeholders removed, verification sign-offs added

Table of Contents

1. Introduction	4
STEP 1 Scope and application	4
STEP 2 Tools and fastening method	4
STEP 3 Component identification	4
STEP 4 Foreign and surplus stock control	4
STEP 5 State model and verification strips	5
STEP 6 Execution-order policy	5
STEP 7 Verification access rule	5
STEP 8 Replacement-part fabrication	5
STEP 9 Procedure selection	5
2. Front chassis assembly	6
STEP 1 Parts required for this chapter	6
STEP 2 Place the base	6
STEP 3 Install the front chassis	6
STEP 4 Install the front chassis pin	6
STEP 5 Front chassis - verification	7
3. Rear chassis assembly	8
STEP 1 Parts required for this chapter	8
STEP 2 Install the rear chassis	8
STEP 3 Install the front-rear chassis pin	8
STEP 4 Install the rear-rear chassis pin	8
STEP 5 Rear chassis - verification	8
4. Front bracket and wing	9
STEP 1 Parts required for this chapter	9
STEP 2 Mount the front bracket	9
STEP 3 Install the front bracket screw	9
5. Wheel assemblies	10
STEP 1 Parts required for this chapter	10
STEP 2 Install the front wheel assembly	10
STEP 3 Install the rear wheel assembly	10
STEP 4 Wheel assemblies - verification	10
6. Final inspection - Procedure A	11

Table of Contents

STEP 1 State verification	11
STEP 2 Fastener and hardware audit	11
STEP 3 Disposition and records	11
7. Procedure B - rear-chassis upgrade	12
STEP 1 Parts and prerequisites	12
STEP 2 Remove the rear wheel assembly	12
STEP 3 Remove both rear chassis pins	12
STEP 4 Remove the long rear chassis	12
STEP 5 Install the short rear chassis	12
STEP 6 Reinstall both rear chassis pins	13
STEP 7 Reinstall the rear wheel assembly	13
STEP 8 Final inspection - Procedure B	13
8. Nonconformance and rework	14
STEP 1 Rework principle	14
STEP 2 Documented error modes and corrective actions	14
STEP 3 Basis for the verification regime	14

1. Introduction

STEP 1 Scope and application

- This manual governs two controlled procedures on the STEMFIE construction-set vehicle: **Procedure A** - full assembly from the base to the completed configuration (state 13), and **Procedure B** - replacement of the long-brace rear chassis with the short-brace rear chassis (state 13 to state 22).
- The vehicle comprises 36 printed parts organized into 11 tracked components. All verification in this manual is performed against the 11-component state model defined in step 5.
- Notation used throughout: orange squares mark work instructions, red marks warnings and known error modes, blue marks informative notes, green marks verification criteria, and magenta circles mark required parts.

STEP 2 Tools and fastening method

- All fastening is performed by hand: push-fit pins and finger-tightened screws and acorn nuts. No tools are required or permitted.
- Do not use pliers, drivers or other tools. Printed PLA bosses can crack under tool torque. Firm finger torque is the specified tightening level for every fastener in this manual.

STEP 3 Component identification

- Parts are color-coded for identification:
 - **White PLA** - braces and beams (base, chassis structures, bracket)
 - **Silver metallic PLA** - pins (4x)
 - **Magenta PLA** - nuts, washers, wing and pulley
 - **Black tough PLA** - wheels
- Before starting, sort all parts by color and verify completeness against the chapter parts lists. Reject damaged or warped parts.

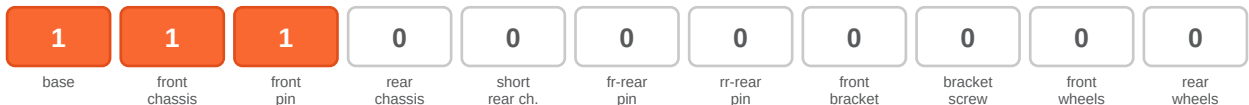
STEP 4 Foreign and surplus stock control

- The workstation contains surplus washers, nuts, screws and pins, including several visually similar pin and washer variants. Not every part at the station belongs in the vehicle.
- Verify the type and length of every fastener against the instruction for that step before installation.
- Rationale: the two highest-frequency documented errors on this build are installation of a wrong pin type and substitution of an acorn nut for the bracket screw. Both are visually subtle and both have escaped automated inspection.

1. Introduction

STEP 5 State model and verification strips

- Throughout this manual, checkpoints display a strip of 11 boxes. Each box represents one tracked component, always in the same fixed order: base, front chassis, front chassis pin, rear chassis, short rear chassis, front-rear pin, rear-rear pin, front bracket, bracket screw, front wheel assembly, rear wheel assembly.
- An orange **1** indicates the component is correctly installed; a white **0** indicates it is not installed.
- At every checkpoint, compare the build to the strip, component by component. Any mismatch is a nonconformance: stop work and proceed per chapter 8.



Example: base, front chassis and front chassis pin installed (state 5)

STEP 6 Execution-order policy

- The procedure is subgoal-oriented. Chapters 2 and 3 may be executed in either order, and where a step states that pin order is interchangeable, either sequence is acceptable. Twenty-two distinct correct execution orders are on record for this build.
- The following are not flexible: every checkpoint must pass before work continues, and a step is complete only when the component is correctly installed and tightened. Execution of an action does not constitute completion of the step.

STEP 7 Verification access rule

- On completing each step, remove both hands from the assembly before verification. This provides an unobstructed view of the build state for the operator, the inspector, or a vision-based verification system.
- Do not build over an unverified or suspect step. A defect concealed beneath subsequent work requires that work to be undone for rectification (chapter 8).

STEP 8 Replacement-part fabrication

- All 36 part geometries are open source (STEMFIE SPS-000001, scaled 200% in all axes). Fabrication parameters: 0.3 mm layer height, 15% infill, 50 mm/s, raft build-plate adhesion, 0.5 mm hole horizontal expansion.
- The 0.5 mm hole horizontal expansion is mandatory; without it the pins will not achieve the specified push fit.

STEP 9 Procedure selection

- Select the applicable procedure from the work order:
- **Procedure A - new build:** proceed to chapter 2 and work through chapter 6 in chapter order (chapters 2 and 3 interchangeable).
- **Procedure B - rear-chassis upgrade:** proceed to chapter 7.
- Procedure B requires an incoming unit verified at state 13 (cover strip). A unit not at state 13 must first be dispositioned per chapter 8.

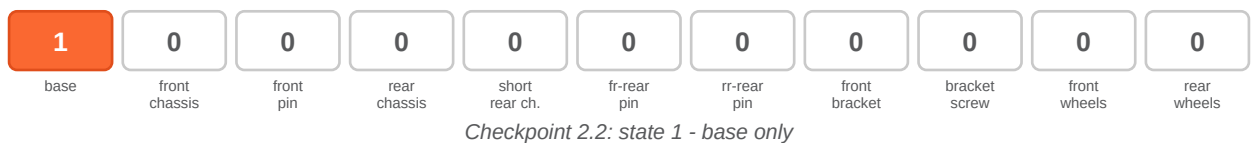
2. Front chassis assembly

STEP 1 Parts required for this chapter

- Base plate (1x) - white
- Front chassis brace set (1x) - white
- Front chassis pin (1x) - silver
- Surplus pins are present at the station. Verify pin type and length against the work instruction before installation (step 4).

STEP 2 Place the base

- Place the base on a flat surface in the standard orientation. All subsequent alignment references the base.
- Inspect the part for damage or warping before proceeding.



STEP 3 Install the front chassis

- Align the front chassis braces to the front interface of the base.
- Fit the chassis onto the base until fully seated: no gap at the joint, bores aligned.
- A misaligned front chassis is a documented high-impact error; all subsequent steps build upon it. Verify squareness at this step.

STEP 4 Install the front chassis pin

- Verify the pin: correct type, correct length, silver.
- Insert the pin through the aligned bores and push to a full seat. Verify head orientation per the work instruction.
- Wrong pin type and reversed pin orientation are the most frequently documented pin errors on this build and have escaped automated inspection. Manual verification is mandatory.

2. Front chassis assembly

STEP 5 Front chassis - verification

- Pin flush and fully engaged; correct type; correct orientation.
- Front chassis cannot rotate or lift from the base.
- Hands clear; build matches the strip below.

1	1	1	0	0	0	0	0	0	0	0
base	front chassis	front pin	rear chassis	short rear ch.	fr-rear pin	rr-rear pin	front bracket	bracket screw	front wheels	rear wheels

Checkpoint 2.5: state 5 - front chassis subgoal complete

Checkpoint 2.5	Operator (name / signature / date)	Inspector (name / signature / date)
Result: PASS [] FAIL []		

3. Rear chassis assembly

STEP 1 Parts required for this chapter

- Long braces (2x) - white
- Front-rear chassis pin (1x) - silver
- Rear-rear chassis pin (1x) - silver
- Use the LONG braces. Short braces are Procedure B stock and must not be installed in a new build. If in doubt, compare brace lengths side by side.
- This chapter may be executed before chapter 2; both orders are correct.

STEP 2 Install the rear chassis

- Align both long braces to the rear interface of the base.
- Fit until the bores align. The braces must sit parallel, bores concentric with the base bores.

STEP 3 Install the front-rear chassis pin

- Verify the pin, then insert it through the forward bore position of the rear chassis.
- Push to a full seat and verify orientation.
- Steps 3 and 4 are interchangeable; the two rear pins may be installed in either order.

STEP 4 Install the rear-rear chassis pin

- Verify the pin, then insert it through the rearmost bore position.
- Push to a full seat and verify orientation.
- Apply the same verification as in chapter 2: wrong type and reversed orientation are the characteristic pin errors. Verify each pin as it is seated.

STEP 5 Rear chassis - verification

- Both pins fully seated, correct type, correct orientation.
- Rear chassis secure with no play; braces parallel.
- Hands clear; with chapter 2 also complete, the build matches the strip below.

1	1	1	1	0	1	1	0	0	0	0
base	front chassis	front pin	rear chassis	short rear ch.	fr-rear pin	rr-rear pin	front bracket	bracket screw	front wheels	rear wheels

Checkpoint 3.5: state 8 - both chassis subgoals complete

Checkpoint 3.5	Operator (name / signature / date)	Inspector (name / signature / date)
Result: PASS [] FAIL []		

4. Front bracket and wing

STEP 1 Parts required for this chapter

- Front bracket (1x) - white, with magenta wing per configuration
- Front bracket screw (1x) - threaded shank with head
- Acorn nuts at the station are visually similar to the bracket screw and are not interchangeable with it. This substitution is a documented error that has escaped automated inspection. Positively identify the screw before installation.
- Chapters 2 and 3 must both be complete before this chapter.

STEP 2 Mount the front bracket

- Align the front bracket to the front chassis interface.
- Fit the bracket and hold the alignment: square to the vehicle centerline, bores aligned.

STEP 3 Install the front bracket screw

- Install the verified screw by hand.
- Tighten to firm finger torque: thread fully engaged, head seated flush.
- HOLD POINT - before proceeding, re-confirm by head shape that the installed fastener is the screw, not an acorn nut.

1	1	1	1	0	1	1	1	1	0	0
base	front chassis	front pin	rear chassis	short rear ch.	fr-rear pin	rr-rear pin	front bracket	bracket screw	front wheels	rear wheels

Checkpoint 4.3 (hold point): state 9 - bracket and screw installed

Checkpoint 4.3 (hold point)	Operator (name / signature / date)	Inspector (name / signature / date)
Result: PASS [] FAIL []		

5. Wheel assemblies

STEP 1 Parts required for this chapter

- Wheels (4x) - black
- Washers (4x) - magenta
- Acorn nuts (4x) - magenta
- Rear pulley (1x) - magenta
- Count the washers before starting. Omission of wheel washers is the documented low-detectability error on this build; the assembly appears acceptable without them.
- Steps 2 and 3 are interchangeable; front and rear wheel assemblies may be installed in either order.

STEP 2 Install the front wheel assembly

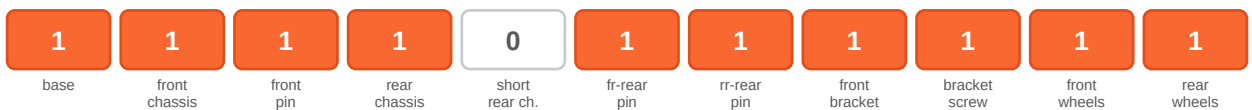
- On each side of the front axle, assemble the stack in order: wheel, washer, acorn nut.
- Install each acorn nut by hand and tighten to firm finger torque.
- Rotation check: both wheels rotate freely with no axial play.

STEP 3 Install the rear wheel assembly

- Install the pulley on the rear axle first.
- A missing rear pulley is a documented error state with low detectability once the wheels are installed. The pulley is always installed before the wheels.
- On each side, assemble wheel, washer, acorn nut, and tighten to firm finger torque.
- Rotation check: both wheels rotate freely; pulley present and seated.

STEP 4 Wheel assemblies - verification

- A washer is present behind every wheel (4x).
- All four acorn nuts tightened to firm finger torque.
- Rear pulley present and seated.
- Hands clear; build matches the strip below if chapter 4 is complete.



Checkpoint 5.4: state 13 - all components installed

Checkpoint 5.4	Operator (name / signature / date)	Inspector (name / signature / date)
Result: PASS [] FAIL []		

6. Final inspection - Procedure A

STEP 1 State verification

- Hands clear. Verify the complete vehicle, component by component, against the strip below.
- Every component at 1 except the short rear chassis, which remains at 0 in the new-build configuration.

1	1	1	1	0	1	1	1	1	1	1
base	front chassis	front pin	rear chassis	short rear ch.	fr-rear pin	rr-rear pin	front bracket	bracket screw	front wheels	rear wheels

Acceptance criterion: state 13 - complete assembly

STEP 2 Fastener and hardware audit

- Front bracket secured by the screw (not a nut), fully tightened.
- All four pins: correct type, correct orientation, fully seated.
- All four acorn nuts tight; washer behind every wheel; rear pulley present; wing present per configuration.
- All four wheels rotate freely; no component exhibits play.

STEP 3 Disposition and records

- Record the build: executed step order, checkpoint results, and the final state vector. On PASS, release the vehicle.
- On any FAIL, do not release. Quarantine the unit and proceed per chapter 8; every component has a sanctioned removal and rework path.

Final inspection - Procedure A	Operator (name / signature / date)	Inspector (name / signature / date)
Result: PASS [] FAIL []		

7. Procedure B - rear-chassis upgrade

STEP 1 Parts and prerequisites

- Completed vehicle, verified at state 13 (cover strip) before work begins
- Short braces (2x) - white, upgrade stock
- Empty, segregated reuse tray (1x)
- This procedure replaces the long rear braces with short braces. The front end of the vehicle is not disturbed: do not loosen any front-end fastener.
- All hardware removed in steps 2-3 is placed in the reuse tray and reinstalled in steps 6-7.

STEP 2 Remove the rear wheel assembly

- Unscrew both rear acorn nuts by hand; remove nuts, washers, wheels and pulley.
- Place all removed hardware in the reuse tray.
- Keep the reuse tray segregated from station surplus stock. Mixing reuse hardware with surplus stock reintroduces the wrong-part error modes of chapter 1, step 4.
- Count-back: 2 wheels, 2 washers, 2 nuts, 1 pulley in the tray.

STEP 3 Remove both rear chassis pins

- Withdraw the front-rear and rear-rear pins straight out, in either order; place them in the reuse tray.
- Withdraw pins axially. Do not lever against the printed bores.
- Both bores clear and undamaged.

STEP 4 Remove the long rear chassis

- Lift the long-brace rear chassis from the base.
- Tag the removed long braces 'removed - long configuration' and route per the work order. They do not enter the reuse tray.
- Hands clear; the stripped rear end matches the strip below and the front end is undisturbed.

1	1	1	0	0	0	0	1	1	1	0
base	front chassis	front pin	rear chassis	short rear ch.	fr-rear pin	rr-rear pin	front bracket	bracket screw	front wheels	rear wheels

Checkpoint 7.4 (hold point): state 18 - rear end stripped, front end intact

Checkpoint 7.4 (hold point)	Operator (name / signature / date)	Inspector (name / signature / date)
Result: PASS [] FAIL []		

STEP 5 Install the short rear chassis

- Verify the braces: compare one against a removed long brace; equal length indicates the wrong part.
- Align and fit the short braces to the rear interface: parallel, bores concentric.

7. Procedure B - rear-chassis upgrade

STEP 6 Reinstall both rear chassis pins

- From the reuse tray, insert each pin to a full seat with correct orientation, in either order.
- Both pins correct type and orientation; chassis secure with no play.

STEP 7 Reinstall the rear wheel assembly

- From the reuse tray: install the pulley first, then wheel, washer and acorn nut on each side. Tighten to firm finger torque.
- Washers and pulley present; nuts tight; wheels rotate freely.
- Reuse tray empty. Any remaining item indicates hardware missing from the vehicle.

STEP 8 Final inspection - Procedure B

- Hands clear. The vehicle matches the strip below: identical to state 13 with the chassis positions exchanged.
- Short braces installed; no long-configuration hardware remaining on the unit; front end undisturbed and tight.
- Record the job: executed step order, checkpoint results, final state vector, and routing of the removed braces. On PASS, release the vehicle; on FAIL, proceed per chapter 8.

1	1	1	0	1	1	1	1	1	1	1
base	front chassis	front pin	rear chassis	short rear ch.	fr-rear pin	rr-rear pin	front bracket	bracket screw	front wheels	rear wheels

Acceptance criterion: state 22 - upgrade complete

Final inspection - Procedure B	Operator (name / signature / date)	Inspector (name / signature / date)
Result: PASS [] FAIL []		

8. Nonconformance and rework

STEP 1 Rework principle

- Every component has a sanctioned removal step. Rework always follows the sequence **remove - verify - reinstall correctly**. Adjustment in place, forcing or bending is not permitted.
- Where a defect is concealed beneath subsequent work, that work must first be undone in the reverse of the legal build order: wheel assemblies before pins, pins before chassis structures.

STEP 2 Documented error modes and corrective actions

- **Washer omitted at a wheel** - remove the affected wheel assembly, rebuild the complete stack, retighten, repeat the rotation check.
- **Wheel omitted** - install with the complete stack; repeat the chapter 6 audit.
- **Wrong pin type or reversed pin** - withdraw the pin, quarantine it from kit stock, install the verified-correct pin, repeat the full-seat check.
- **Acorn nut installed in place of the bracket screw** - remove the nut, quarantine it, install the screw, tighten, repeat the chapter 4 hold point.
- **Rear pulley omitted** - remove the rear wheel assembly, install the pulley, rebuild per chapter 5, step 3.
- **Wrong braces installed (long/short)** - strip the rear end per chapter 7, steps 2-4; fit the correct braces; rebuild per chapter 7, steps 5-7.
- **Omitted step discovered late** - establish the true build state against the strips; undo the minimum legal set of steps to expose the omission; complete it; rebuild, repeating every checkpoint on the return path.

STEP 3 Basis for the verification regime

- In recorded executions of this build, 42% of runs contained at least one missing or incorrectly completed step, including runs performed with correct instructions in hand. Several of the documented error modes additionally escaped automated visual inspection.
- The checkpoints, hold points and sign-offs in this manual are derived from those documented failures and are mandatory.